

Bobby Hodgkinson

Associate Teaching Professor | Engineering Education | Experiential and AI-Integrated Learning

825 Waite Drive, Boulder, Colorado 80303 • 352.275.9477 • hodgkinr@colorado.edu

bobpe.com • [LinkedIn](#)

Professional Profile

Associate Teaching Professor in the Ann and H. J. Smead Department of Aerospace Engineering Sciences at the University of Colorado Boulder. Designs experiential, mastery-oriented, and AI-integrated learning environments that emphasize engineering judgment, verification and validation, student agency, and timely individualized feedback. Teaching and curriculum work spans introductory computing, large-enrollment aerospace laboratories, experimental methods, project-based learning, and senior capstone.

Academic and Professional Experience

Associate Teaching Professor

Fall 2024–Present

University of Colorado Boulder, Smead Aerospace Engineering Sciences

Leads course and curriculum development, experiential laboratory education, and responsible uses of AI for teaching and feedback. Mentors undergraduate researchers, learning assistants, and student developers.

Consultant

June 2026–Present

Blackboard

Contributes to development of the Scholar Learning Management System, bringing experience in learning-platform design, instructional workflows, and AI-integrated education.

Assistant Teaching Professor

August 2018–Fall 2024

University of Colorado Boulder, Smead Aerospace Engineering Sciences

Taught and coordinated high-enrollment aerospace laboratory courses; developed experimental hardware, assignments, assessment workflows, and instructional teams across thermodynamics, aerodynamics, structures, dynamics, controls, computing, and design.

Professional Research Assistant

July 2013–August 2018

University of Colorado Boulder, Smead Aerospace Engineering Sciences

Provided instrumentation and electronics support for teaching laboratories, capstone teams, graduate projects, and research. Designed and validated experimental hardware and software aligned with course learning objectives.

Laboratory Manager

May 2011–May 2013

University of Florida, Institute for Networked Autonomous Systems

Managed personnel, electronics and machine shops, and improvements to aerial, surface, and underwater robots. Advised undergraduates preparing for the MATE ROV competition.

Education

M.S., Aerospace Engineering, University of Colorado Boulder, 2011

Focus: Vehicle Controls

B.S., Aerospace Engineering, University of Colorado Boulder, 2010

Minor: Electrical Engineering

Teaching and Curriculum Leadership

- **ASEN 3501, Experimental Methods (Fall 2026).** Led extensive development and first delivery of a three-phase, six-outcome course organized around execution and uncertainty quantification, predictive modeling and comparative validation, and experimental design with verification and validation planning. Built a mastery-evidence structure using oral assessment, comparative critique, peer evaluation, and a transparent assessment-to-outcome ontology. Created a [public shared-source course environment](#) embedded in Canvas so collaborators and students can inspect the complete course sequence and outcome alignment while content is maintained in one location.
- **ASEN 1030, Introduction to Computing for Aerospace Engineering (Fall 2025).** Designed and taught an inquiry-based computing course centered on program comprehension, explanation, code interviews, MATLAB Academy credentials, and human-reviewed AI feedback. The FCQ record lists 51 students in Bobby's section; the shared feedback workflow served approximately 100 students across both course sections and processed 969 submissions. The human-reviewed workflow returned individualized feedback within 24 hours; reported median review time was under one minute per response, and total inference cost was approximately \$7.50 for the term.
- **Canvas-integrated peer-evaluation feedback system.** Developed an end-to-end system that compiles peer-evaluation comments, synthesizes individualized constructive feedback, returns feedback to students, and provides

instructors with a summary that flags notable comments or patterns.

- **ASEN 2802, Aerospace Sciences Laboratory I (2022–2025).** Led repeated delivery and substantial redevelopment of large-enrollment aerodynamics, structures, and thermodynamics laboratory experiences, including balloon experiments and a combined aeroelasticity activity. Required students to compare analytical, experimental, computational, and AI-generated results and defend discrepancies.
- **ASEN 2002/2003 and ASEN 2803, Sophomore Dynamics and Laboratory Sequence (2018–2025).** Coordinated multi-section instruction, hardware demonstrations, MATLAB-based analysis, experimental methods, and iterative assessment improvements across thermodynamics, aerodynamics, dynamics, and controls.
- **ASEN 4018/4028, Senior Projects (2013–Present).** Delivered technical workshops and supported project definition, instrumentation, fabrication, software, testing, design review, and evaluation through the Project Advisory Board and formal course roles.
- **ASEN 2012 and GEEN 1400.** Taught Experimental and Computational Methods in Aerospace Engineering and first-year Engineering Projects lecture/laboratory offerings, including approximately 30 students in ASEN 2012 and 20–30 students in the GEEN 1400 offerings.
- **ASEN 2519, Aerospace CAD/CAM Basics (Maymester 2014 and 2015).** Team-taught SolidWorks, Solid-CAM, aerospace machining, manufacturing safety, design practices, and shop techniques through 18 lecture-style sessions in each four-week course for 15–20 students.
- **Laboratory experiments and instructional support (2013–Present).** Delivered recurring introductions and small-group instruction for laboratory activities in statics, structures, thermodynamics, aerodynamics, dynamics, vehicle design, and electronics, often addressing 50–80 students at a time. Provided additional instructional support across ASEN 2502, 3128, 3200, 3801, 3802, and 5114.

Course Enrollment Record

Enrollment is summed across all sections for each course and term. The record contains **3,739 student-course enrollments across 24 term-course offerings** from Fall 2020 through current Fall 2026 enrollment. The FCQ workbook documents 78 sections through Spring 2026; Fall 2026 figures are current candidate-reported enrollments.

Term	Course	Course title	Enrollment
Fall 2026	ASEN 3501	Aerospace Experimental Methods (current)	176
Fall 2026	ASEN 1030	Introduction to Computing for Aerospace Engineering (current)	73
<i>Spring 2026</i>	<i>ASEN 4028</i>	<i>Senior Project 2: Design Practicum</i>	<i>230</i>
Fall 2025	ASEN 1030	Introduction to Computing for Aerospace Engineering	51
<i>Fall 2025</i>	<i>ASEN 4018</i>	<i>Senior Project 1: Design Synthesis</i>	<i>230</i>
Summer 2025	ASEN 2803	Dynamics and Controls Laboratory	33
Spring 2025	ASEN 2802	Aerospace Sciences Laboratory I	35
<i>Spring 2025</i>	<i>ASEN 4028</i>	<i>Senior Project 2: Design Practicum</i>	<i>221</i>
Fall 2024	ASEN 2802	Aerospace Sciences Laboratory I	311
Summer 2024	ASEN 2803	Dynamics and Controls Laboratory	28
Spring 2024	GEEN 1400	Engineering Projects	30
<i>Spring 2024</i>	<i>ASEN 4028</i>	<i>Senior Project 2: Design Practicum</i>	<i>214</i>
Fall 2023	ASEN 2802	Aerospace Sciences Laboratory I	252
<i>Fall 2023</i>	<i>ASEN 4018</i>	<i>Senior Project 1: Design Synthesis</i>	<i>214</i>
Summer 2023	ASEN 2803	Dynamics and Controls Laboratory	22
Spring 2023	ASEN 2802	Aerospace Sciences Laboratory I	20
Spring 2023	ASEN 2803	Dynamics and Controls Laboratory	122
Fall 2022	ASEN 2802	Aerospace Sciences Laboratory I	273
Spring 2022	ASEN 2003	Dynamics and Systems - Lab	264
Spring 2022	ASEN 2012	Experimental/Computational Methods in Aerospace Engineering	31
Fall 2021	ASEN 2002	Thermodynamics and Aerodynamics - Lab	313
Spring 2021	ASEN 2003	Dynamics and Systems - Lab	262
Spring 2021	GEEN 1400	Engineering Projects	27
Fall 2020	ASEN 2002	Thermodynamics and Aerodynamics - Lab	307
Course enrollments excluding senior-project support			2,630
All enrollments including senior-project support			3,739

Italicized rows are ASEN 4018/4028 senior-project support assignments. They are retained as a distinct form of teaching and are excluded from the first subtotal. Fall 2026 values are current enrollments and may change before the end of the term.

Coursera Courses and Learner Reach

Creator and instructor of a three-course graduate sequence in generative AI as part of the CU MS-AI, MS-CS, and MS-DS programs. Learner totals combine public and private course enrollments.

Course	Learners	Link
Introduction to Generative AI (January 12, 2026–Present)	2,714	Course page
Modern Applications of Generative AI (May 4, 2026–Present)	1,020	Course page
Advances in Generative AI (May 4, 2026–Present)	858	Course page
Total learner-course enrollments	4,592	

Publications and Conference Papers

Engineering education and AI-integrated learning

- **Hodgkinson, B.** “AI-Integrated Inquiry-Based Computing Course.” ASEE Annual Conference, 2026, Paper 50826.
- **Hodgkinson, B.** “An Aeroelasticity Experiential Learning Activity with AI-Driven Comparative Analysis and Evaluation.” 2025 ASEE Annual Conference & Exposition, Paper 57598.
- Long, C., and **Hodgkinson, B.** “Evaluating LLM Performance for Generation of Concept Assessment Questions and Individualized Student Feedback in Aerospace Engineering Sophomore Lab.” 2025 ASEE Rocky Mountain Section Conference, Paper 55220.
- Whittenburg, N., and **Hodgkinson, B.** “Transcription-Enhanced AI Code Review for Deeper Learning.” 2025 ASEE Rocky Mountain Section Conference, Lightning Talk 44.
- **Hodgkinson, B.**, and Whittenburg, N. “WIP: AI-Based Sentiment Analysis and Grader Enhancements.” 2024 ASEE Annual Conference, Paper 42183.
- **Hodgkinson, B.** “WIP: Towards an AI Teaching Assistant for Aerospace Engineering Lab Courses.” 2024 ASEE Rocky Mountain Section Conference, Paper 49.
- **Hodgkinson, B.**, and Schwartz, T. “BYOE: An Apparatus for Exploring Small-Satellite Estimation and Control.” 2021 ASEE Annual Conference, Paper 32904.

Aerospace systems, robotics, and sensing

- **Hodgkinson, B.**, Lipinski, D., Peng, L., and Mohseni, K. “High Resolution Atmospheric Sensing Using UAVs.” *Distributed Autonomous Robotic Systems*, vol. 104, pp. 31–45, Springer, 2014. DOI: 10.1007/978-3-642-55146-8.
- **Hodgkinson, B.**, Lipinski, D., Peng, L., and Mohseni, K. “Cooperative Control Using Data-Driven Feedback for Mobile Sensors.” IEEE International Conference on Robotics and Automation, 2013.
- Peng, L., Lipinski, D., **Hodgkinson, B.**, and Mohseni, K. “Dynamic Data Driven Application System for Puff Estimation Using UAVs.” International Conference on Unmanned Aircraft Systems, 2013.
- **Hodgkinson, B.**, Shyu, D., and Mohseni, K. “Acoustic Source Localization System Using a Linear Arrangement of Receivers for Small Unmanned Underwater Vehicle.” IEEE/MTS OCEANS, 2012.
- **Hodgkinson, B.**, Lipinski, D., Peng, L., and Mohseni, K. “High Resolution Atmospheric Sensing Using UAV Swarms.” International Symposium on Distributed Autonomous Robotic Systems, 2012.
- **Hodgkinson, B.**, Lipinski, D., Peng, L., and Mohseni, K. “Environmental Monitoring with K-Means Error Reduction Using UAVs Controlled by a Fluid-Based Scheme.” IEEE/RSJ IROS Workshop on Robotics for Environmental Monitoring, 2012.
- Krieg, M., Klein, P., **Hodgkinson, R.**, and Mohseni, K. “A Hybrid Class Underwater Vehicle: Bioinspired Propulsion, Embedded System, and Acoustic Communication and Localization System.” Marine Technology Society, pp. 153–164.
- Ambrosio, D., Gizzi, R. D., **Hodgkinson, B.**, Kirkpatrick, J., Miller, C., Price, J., and Thomas, T. “CUBOAT: Colorado Underwater Buoyant Oceanic Acoustic Network.” 49th AIAA Aerospace Sciences Meeting, 2011.

Invited Talks and Other Conference Presentations

- Kiewit Faculty Open House, invited national presentation translating AI-in-education deployments for approximately three dozen civil-engineering and construction-management faculty, June 8–10, 2026.
- MathWorks Educator Conference Debrief, Boulder, Colorado, January 28, 2026.
- MathWorks Educator Conference, Carleton College, October 25, 2025.
- RMACC AI Workshop, August 5, 2025.
- AI @ CU Boulder Showcase, Boulder, Colorado, April 18, 2025.

- NAIRR Pilot Program Talk, April 2, 2025.
- 2025 ASEE Annual Conference & Exposition.
- 2025 ASEE Rocky Mountain Section Conference, University of Colorado Boulder, May 14–16, 2025.
- ASEE Annual Conference, Portland, Oregon, June 23–26, 2024.
- ASEE Rocky Mountain Section Conference, Boulder, Colorado, May 15–18, 2024.
- CU–KAIST educational collaboration, Daejeon, South Korea; represented CU and presented/shared aerospace spin-module experiential-learning work, May 14–18, 2023.
- ASEE Annual Conference, virtual meeting, July 26–29, 2021.
- IEEE/MTS OCEANS, Hampton Roads, Virginia, October 14–19, 2012.
- AIAA International Student Conference, Orlando, Florida, January 4–7, 2011.
- National Instruments Student Design Showcase, Austin, Texas, August 3–5, 2010 (first place).

Leadership and Service

Department and College Leadership

- Chaired teaching-faculty search work in 2022–2023, coordinating application review, interviews, faculty presentation, and recommendation. The 2023 search reviewed more than 30 applications and concluded with a successful hire.
- Sustained educational laboratory infrastructure serving multiple Smead courses, including 24 electronics workstations, software and supplies, vendor relationships, staff transitions, and onboarding needed for instructional continuity.
- Served on working groups for instrumentation, dynamics, thermal/fluid systems, undergraduate curriculum, quality assessment, lab scheduling, and the new curriculum across multiple academic years.
- Served on the ITLP Governance Committee and Teaching Professor search work; contributed to CEAS generative-AI strategy.
- Faculty advisor for Hyperloop and Sigma Gamma Tau; provided recurring design, project-management, technical-review, academic, and career guidance.
- Dean's ad hoc budget committee member; advised the College of Engineering dean on budget recommendations responding to COVID-19 funding reductions.
- Engineering Social Club organizer; coordinated social events for College of Engineering students and faculty.
- Undergraduate Committee member; contributed to strategies for improving and maintaining the aerospace engineering curriculum.
- AeroBEER Engineering Education Research Group participant; reviewed engineering-education scholarship and discussed research-based improvements to instruction.
- Computing Committee member; provided curriculum and instructional-computing recommendations.
- Capstone COVID Workgroup member; developed cross-college strategies and outcome expectations for senior design courses during the pandemic.
- Long-running Project Advisory Board service for senior capstone, including presentation review, critique, questioning, and evaluation.

Campus and CU System Leadership

- Co-developed separate student and faculty/staff responsible-use courses required for access to [ChatGPT EDU](#), working with approximately six colleagues across CU Boulder and CU Denver. Contributed technical guidance, curriculum structure, and AI-literacy content.
- Academic Technology Advisory Group member; contributed to university AI and educational-technology strategy, participated in a small AI/design-based education research working group, and brought Australian AI-literacy practices into CU discussions and personal teaching through professional exchange with Danny Liu.
- Center for Teaching & Learning AI Literacy Ambassador (Spring 2025); delivered and supported campus and college faculty development on responsible generative-AI use, including practical AI-literacy and NotebookLM programming.
- Conversations on Generative AI participant and presenter; engaged in weekly College of Engineering discussions about generative-AI developments and institutional directions.
- Participated in a CTL Students as Partners exploration with a graduate theater student, identifying student interests and mapping existing CU courses relevant to a possible undergraduate AI certificate.
- Completed the 2025 Online Teaching Academy and applied its online pedagogy and Canvas course-design practices to ASEN 2803.

Professional and Community Leadership

- Rocky Mountain AI Interest Group executive-board member and founder/lead of the AI in Education subgroup; organized recurring educator and community programming and helped connect regional practitioners, educators, and public audiences around AI literacy.
- Served on the St. Vrain Valley School District Innovation Center AI Advisory Board and a Boulder Valley School District AI advisory committee, contributing a higher-education perspective on AI literacy and education.
- Served on the 2025 ASEE Rocky Mountain Section conference planning team and reviewed submissions for ASEE national and regional conferences.
- Make4Covid contributor; fabricated and distributed personal protective equipment during the initial COVID-19 supply shortage.

Awards, Fellowships, and Project Support

- CU System AI Recognition Award, AI for Teaching & Learning, 2026.
- CU Innovation & Efficiency Award, Top Finalist for *AI-Augmented Personalized Feedback*, 2026.
- Center for Teaching & Learning Excellence in Teaching with Technology Award nomination.
- Dean's Excellence Fund Fellowship supporting undergraduate research through the BUFFALOES AI-literacy initiative.
- National AI Research Resource / Texas Advanced Computing Center allocation supporting educational knowledge-graph and retrieval-augmented-generation development.
- Advanced a classroom knowledge-graph and AI-support suite to round two of CU Boulder's Embark program (2025).
- Invited by PI Tim Minton to serve as educational director on a MISSE proposal and developed its education component; proposal not funded.
- National Instruments Student Design Showcase, first place, 2010.

Mentoring and Faculty/Student Development

- Supervised and trained learning-assistant teams supporting aerospace laboratory instruction, course development, equipment validation, safety, and student feedback.
- Mentored four student developers building a classroom clicker platform, knowledge-graph tutoring chatbot, Canvas-derived study guides, and a Canvas-integrated chatbot. Advised two SPUR/UROP researchers on reinforcement-prompting and image-recognition projects; one research contribution was incorporated into the ASEN 1030 feedback pipeline.
- Mentored teaching-faculty colleague Charlie Hoke and participated in ACUMent, supporting peer development in teaching and academic practice.
- Mentored two CU Science Discovery high-school students investigating when frontier chatbots help or hinder novice learners completing an advanced aerospace aeroelasticity activity; the students completed and presented their work.
- Mentored aerospace students through Presidents Leadership Class and other academic and professional-development programs.

Community Outreach and Public Education

- Designed and delivered practical AI-literacy workshops for Parent Engagement Network and the Louisville YMCA, including *Navigating AI: Build Your Parent Playbook* (November 2025) and an April 2026 guided workshop in which parents explored ChatGPT, Claude, and NotebookLM.
- Delivered additional AI-in-society and AI-literacy programs for Balfour retirement communities, Boettcher Scholars, Naropa faculty, Women of Aeronautics and Astronautics, Front Range Community College, and other public audiences.

Online Education, Media, and Public Engagement

- Guest, *Smarter Campus Podcast*: "Engineering Understanding: AI, Aerospace, and Rethinking Learning," July 21, 2025.
- Guest, *Parenting Well Podcast*: "Raising Thinkers in the Age of AI," January 7, 2026.
- Guest, *Conscious Millionaire Podcast*: "Personalized Solutions for Science and Tech Customers."
- Featured by CU Boulder in "Adding AI to the Aerospace Classroom."
- Featured in *Colorado AI News*, "Pic of the Week: CU Boulder professor Bobby Hodgkinson brings AI awareness to the senior class," March 22, 2025.